

Part 2

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MALAWI SCHOOL CERTIFICATE OF EDUCATION

MSCE

AGRICULTURE
SUMMARY NOTES

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UNIT 1.

SOIL DEGRADATION

Soil degradation refers to a loss in the value and quality of the soil.

It is caused by erosion and a loss of soil fertility.

The rate of soil erosion is accelerated through human activities including agricultural operations. Vegetation is cleared, leaving the soil bare, exposing it to wind and water. The result is chemical, biological and physical degradation.

Soil degradation also results in an accumulation of salts and loss of soil fertility.

Causes of soil erosion and degradation

- (a) Water Erosion
 - Splash erosion
 - Rill erosion
 - Gully erosion.
 - Sheet erosion
- (b) Wind erosion : soil is dried and is blown away.
- (c) Physical erosion occurs when structure of soil is destroyed by rain or machines. If the structure is destroyed by rain, the soil particles may not be carried away, instead the soil becomes puddle – this is known as Puddle erosion.
- (d) Chemical degradation
 - Excessive use of inorganic fertilizers (e.g. Sulphate of Ammonia) and pesticides result in chemicals degradation – may make soil become acid.
- (e) Biological degradation
 - People trampling as they walk about.
 - Termites may destroy soil Structure
- (f) Excess salts
 - Salts build up in the soil when there is poor drainage and low rainfall, when fertilizers are applied and the soil is irrigated with poor water.
- (g) Loss of soil fertility
 - Through wind and water erosion.
 - Through continuous cropping.

Some bad farming practices that cause soil degradation include: -

- Cultivating steep slopes, river or stream banks.
- Constructing ridges along the slopes.
- Use of heavy machines.
- Continuous cropping.
- Cultivating the soil when it is too dry or wet.
- Wanton cutting down of trees – deforestation.
- Uncontrolled fires.
- Overgrazing.
- High human population.

Effects of soil degradation

- Loss of soil fertility, so less yield.
- Reduced infiltration and increased run-off.
- Rivers, dams, lakes sitting up due to sedimentation.
- Rivers drying up.
- Low ground water table.
- Flooding
- Loss of arable land.
- Loss of grazing land.

Control of soil degradation

- Biological i.e. provide vegetable soil cover.
 - Crop rotation
 - Mulch
 - Controlled grazing.
- Physical
 - Construct storm drains, box or tie ridges, contour bunds.

UNIT 2

CROP IMPROVEMENT

Is increasing the production of cultivated plants by developing better cultivars (shortened form of cultivated varieties)

This is done by plant breeders. The plant breeder identifies plants with desirable characteristics and develops them to increase the productivity of the crop.

The performance or productivity of any crop depends on the inheritable traits the crop has, as well as the environmental conditions under which it is grown.

Aims and objectives of crop improvement.

- To increase the average crop yield (yield quantity)
- To improve the quality of the produce (yield quality)

Objectives (specific aims) that relate to increasing yield quantity include:-

- To increase the biomass (the capacity to produce and retain an adequate quantity of dry matter or vegetable matter).
- To improve partition (the capacity to divert biomass to the desired harvestable portion of the plant) : a higher quantity of crop yield may be obtained, not because of increased plant biomass, but because during photosynthesis the plant is able to direct most of the manufactured food to the harvestable part of the plant such as the root (root tubers, stem (sugarcane or timbers) leaves (cabbage) etc.
- To increase resistance to pests and diseases.
- To improve seasonal adaptation..
- To increase tolerance to adverse environment conditions such as drought, hail, cold, heat.

Objectives that relate to increasing yield quality include

- To achieve uniformity (having the same characteristics between individual crop plants in the field.)
- To increase the nutritive value of crop products.
- To promote 'dwarf ness' in crop: this an important breeding objectives in fruits because it makes harvesting quicker, cheaper and more thorough. In rice it reduces lodging.
- To improve processing qualities of crop products. Buyers may want maize which can give more white flour (milling quality); high oil content (ground nut) ; sugar content e
- To improve / increase market value of crop products by developing products of better colour, texture and taste (palatability).

Methods of crop improvement

- Introduction
- Selection
- Hybridization (cross breeding).

(i) Introduction

This is the importation of crop varieties of superior productivity to serve as foundation stocks for breeding or propagation usually after quarantine and testing. For instance:

Maize : (UCA) from Tanzania

Rice : (Blue Bonnet) from USA

Groundnuts : Chitedze groundnuts (CG7)

(ii) Crop improvement through selection

Choosing a superior plant that has desirable characteristics for propagation or breeding remember (like begets like) e.g.

- Chalimbana groundnuts from Zambia, was chosen through single plant Selection.

- Faya 14-m – 69 (rice) is a selection from several strains of faya rice.

Several ways of selecting plants for breeding and some of them are:-

Mass selection

This means choosing a large number of plants showing desirable characteristics, for seed.

These imported plants are grown and the law of 'Survival of the fittest' (natural selection) is allowed to operate: those plants not Favoured in the new environment die off.

The breeder then eliminates (out of the survivors) plants that are obviously inferior.

Seeds from the rest of the survivors is mixed (bulked) and used for the next cycle of selection

Refer notes on page 58

UNIT 3.

IMPORTANCE AND TYPES OF PASTURES.

Pasture is the cheapest and largest source of feed for ruminants.

Importance of Pastures

- (a) Pasture is grassland used for grazing. It may also defined as grass and / or other herbaceous plants grown and used as feed by grazing livestock.
- (b) Improves and maintain soil fertility.
 - Through circulation of nutrients pasture eaten / dung and urine.
 - Some pastures are leguminous – they help to add nitrogen to soil.

Value of nitrogen fixation by some pasture legumes is shown below

Country	Legume	N-fixed kg/ha
Malawi	Silver leaf	87
Malawi	Siratro	74
Zimbabwe	Lotononis	62
Kenya	Silver leaf	178
Uganda	Stylo	161

- (c) Pastures improve soil structure through organic matter decomposition.
- (d) Pasture protects soil from soil erosion through:

- Fibrous root system.
 - Provision of soil cover.
- (e) Pastures control pests and diseases. Love grass and Katambora Rhodes grass are used in rotations of tobacco to control eelworms

These grasses are resistant to eelworms

Types of pastures

a. Natural Pastures

Are uncultivated grassland (dry or dambo) in which indigenous species are dominant

b Cultivated Pastures

These are two types, namely

- (i) Permanent pasture. These are managed for 10 – 15 years eg star grass (*cynodon dactylon*) buffer
- (ii) Temporary pasture (also called leys) are managed for 3 – 5 years eg. Rhodes grass, guinea grass, Napier grass and setaria.

They may consist of a pure stand of grass or legume or the interplanting of the two.

Benefits of cultivated pastures (improved) over natural pastures

- High dry matter yield eg Siratro 9,655 tons/ha
Guinea grass 10,075 tons/ha

While natural pasture such as

unimproved dambo pasture 3,220 tons / ha.

dry land 1,050 tons / ha.

- Fixed nitrogen by legumes improves soil.
- Higher protein content.
- Higher digestibility..
- Legumes retain high nutritive value even when mature.

UNIT 4

PASTURE ESTABLISHMENT

This involves sowing pasture correctly, following recommended methods and procedures.

Methods of sowing/ establishing pasture

(a) Broadcasting

Advantages

- Easiest way of sowing tiny seeds e.g. less labour required to sow.
- Quickest way of sowing.

Disadvantages

- Some seeds may be picked up by birds or washed away by rainwater.
- Some seeds may be buried too deep and fail to germinate.

- Uneven seed distribution, resulting in overcrowding.
- Requires a lot of seed, so it is expensive.
- Difficult to carry out operations such as weeding, fertilizer application.

(B) Drilling

Seed is sown in trenches (drills) made by sticks, hoes or machinery and then covered by soil.

Advantages

- Seeds are placed at desired depth and distance.
- Easy to carry out operations. E.g. applying fertilizer.

Disadvantages

- May be time consuming if done manually
- It may be expensive if done mechanically

(C) Oversowing (Sod seeding)

Is sowing seed (pasture) into an already established pasture. For effective establishment, the existing pasture species need to be reduced by burning or heavy (severe) grazing before sowing.

Advantages

- Improves composition of existing pasture since desired species are introduced.
- Increases the dry matter yield of natural pastures (unimproved)
- Increases quality of natural pasture in terms of protein content.

Disadvantages

- The introduced pasture may face competition from the already established pasture.

(D) Undersowing

Is sowing pasture seed under a growing arable crop such as maize (when it 45 cm high).

Advantages

- Pasture become established at least a year earlier (ie instead of waiting to harvest the arable crop first)
- Enable farmer to harvest an extra crop of maize from the plot.
- Eliminates the cost of land preparation for the pasture crop.
- Young pasture crop is protected from adverse weather e.g. sun scorch or strong winds.
- Increased total (combined) yield from the plot.

Disadvantages

- The yield of each crop (arable and pasture crop) is less than if it were direct planted (in pure stand)

- Competition for nutrients lights. Etc

(E) Vegetative planting (vegetative propagation)

- Stem cuttings (setts, root cuttings (splits) are used. For example: some legumes can be planted vegetatively such as lotononics variety (stem or roots are used) Silver leaf stolons are used.

Advantages

- Plants grow and mature faster
- Problems of dormancy is reduced, no need of clarification (seed treatment by eg boiling)
- Ensure genetic uniformity
- Is the only way of propagating crops whose seeds are not viable e.g. bananas.
- Vegetative organs e.g. stems are more hardy than seedlings i.e. they are able to withstand environment hazards (such as drought) better.
- They (vegetative parts) have sufficient food reserves to help the young shoots to become established.
- Vegetative planting materials are readily available.

Disadvantages

- Higher risk of transferring diseases to new plants.
- It is more difficult to introduce variation into the crop, making crop improvement difficult.
- Vegetative planting material tend to be bulky so difficult to handle, transport.
- Some of the procedures in vegetative propagation require specialist knowledge and skill (e.g layering, grafting or budding) and are slow and tedious.

Pasture establishment procedure.

These procedures include the following:

- Choosing right time for sowing.
- Preparing the land.
- Selecting good quality seed.
- Selecting the appropriate type of pasture grasses and legumes.
- Selecting the right amount of seed per hectare..
- Processing the seeds for successful germination.
- selecting an effective sowing method.
- Applying fertilizer.

(I) Time for sowing

Plant early, not later than end of January.

(II) Land preparation

Clear land plough early to expose perennial weed seeds to sun, and crop residues to decompose. Large clods should be broken down to ensure fine tilth. Note on steep slopes fine tilth encourages may encourage run-off and soil erosion.

(III) Selecting pasture seed

Successful pasture establishment depends on the quality of the seed. Seed quality is expressed as pure line seed content (PLSC). It is calculated as follows:

$$\text{PLSC} = \frac{\text{Purity \%} \times \text{Germination \%}}{100}$$

Quality seed is one with:

- High germination %
- Pure (not mixed with other pasture seeds or other crops)
- No trash, weed seeds, diseases

Note: Use certified seed

(IV) Selecting appropriate pasture grasses and legumes

Consider the following factors:-

- Adaptability to the environment (soil climatic conditions)
- High overall dry matter production per unit area (i.e high yield). Pasture should provide two or three cuts or grazing per year and should be capable of regenerating (i.e. regrow)
- High feeding value. I.e. high crude protein. Feeding value of some pastures decline after flowering.
- Proposed duration of pasture – Is it a temporary or permanent pasture? Some pastures e.g. Rhodes grass are suitable only as leys because their productivity declines markedly after four years, unless the sward is very well managed (see table below)
- Compatibility with desired species.

Choose grasses or legumes that can successfully grow together with other chosen species since a grass / legumes mixture gives better results than pure stands. Climbing legumes (Siratro and silver leaf) are selected for growing with taller grasses, while shorter legumes (stylo and Lotononis) are selected for growing with the shorter grasses. (See table below)

- Proposed method of utilization of the pasture.

The choice of pasture grasses or legumes should depend on whether they will be used for grazing or hay, silage or cut and carry. (green fodder or silage). The ease of cutting and handling the fodder is an important factor to consider Leucaena is suited for cutting and carry while lotononis and stylo are not. (See table below)

RECOMMENDED UTILISATION OF GRASS

Grasses	Compatible legumes	Utilisation		
		Grazing	Hay	Cut & carry
Annual forages				